

Tejas Arya

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SUMMARY

Applied Scientist (M.Tech in AI & ML, IIT Hyderabad) specializing in **Agentic AI**, **Generative AI**, and **Machine Learning** — architecting production-grade **Multi-Agent LLM systems**, **RAG** pipelines, recommendation engines, and **MLOps** infrastructure processing **10,000+** customer interactions daily.

PROFESSIONAL EXPERIENCE

Tekion Corp
Applied Scientist I

Bengaluru
Jul 2024 – Present

Sales Bot — multi-agent conversational commerce platform

- Architected an **Agentic AI** platform — a **Multi-Agent Conversational AI** built on **LangGraph** — automating inbound dealership SMS end-to-end across vehicle discovery, recommendations, and transaction routing.
- Engineered dynamic **Tool Calling** over REST APIs (appointment booking, live dealership hours) and a **ReAct Agent** routing natural-language vehicle search across **Elasticsearch**, **Vector DBs**, and **Knowledge Graphs**.
- Designed a **Knowledge Graph** state store linking vehicle inventory to lead context, powering a **Graph Embeddings** recommendation engine that surfaces high-relevance alternatives.
- Scaled an LLM-driven **Call Intent Classification** microservice in **Python** (**scikit-learn** baselines, **Docker**, **Kafka**) to **10,000+ transcripts/day** with automated backpressure handling.
- Instrumented **OpenTelemetry** distributed tracing for real-time monitoring of agent interactions, eliminating LLM inference latency bottlenecks in production.
- Established an **MLflow on Databricks** MLOps pipeline to continually monitor model drift, response precision/recall, and API reliability.

CRM Pro — generative AI email assistance at dealership scale

- Built a generative AI **Email Suggestion Engine** serving **5,000+ emails/day** — **30%** of total dealership email volume and **\$6M** in attributable annual revenue.
- Engineered context-aware topic ranking over CRM history plus a **multi-channel summarization pipeline** (calls, emails, chats) to build high-signal prompt context for personalized, compliance-aligned drafts.
- Optimized real-time latency via I/O profiling and **Prompt Engineering**, orchestrating asynchronous **Kafka** event flows with versioned context in **MongoDB** and **Redis** caching for zero-downtime generation.

Carelon Global Solutions
Associate Software Engineer III

Hyderabad
Aug 2023 – Jul 2024

- Architected **Python-based Machine Learning** and NLP pipelines (**scikit-learn**, **PyTorch**) for healthcare analytics: **TrOCR** for handwritten text recognition (**~95% accuracy**) and **LayoutLMv3** to boost NER performance by **20%**.
- Fine-tuned **BERT** for domain-specific Document QA (**~85% accuracy**) and integrated a **Vector DB** to slash semantic search query latency by **40%**.
- Deployed a scalable **LangChain** RAG-based QA system, automating the processing of **10,000+** complex medical documents daily.
- Built an automated **OpenCV** text detection pipeline (500+ documents/hour) paired with interactive **Streamlit** dashboards for stakeholder data exploration.

PUBLICATIONS

Vehicle Speed Prediction using Deep Learning 🔗

IEEE, 2024

- Designed a **CNN-GRU** hybrid architecture, extended with a **Transformer** forecaster for temporal stability, fusing wheel-speed, acceleration, and steering-angle signals with ablations on window sizes and sensor subsets.
- Sponsored M.Tech thesis with **Hyundai**: aligned deep learning model interfaces to ABS hardware integration constraints.

MACHINE LEARNING PROJECTS

Machine Learning-Based Clarification Question Generation 🔗

Mar 2022 – Jun 2023

Course Project (Deep Learning) – IIT Hyderabad

- Framed clarification question generation as **sequence-to-sequence** learning; fine-tuned **BART** on the Amazon **QuAC** dataset with systematic **hyperparameter tuning** (LR schedules, batch sizes, beam search).
- Benchmarked variants on **BLEU** and **human evaluations**, adding a **schema-mapping** module that detects missing context to guide follow-up questioning.

EDUCATION

M.Tech, Artificial Intelligence & Machine Learning — Indian Institute of Technology Hyderabad

Aug 2021 – May 2023

Bachelor of Engineering — Netaji Subhas Institute of Technology, Delhi

Aug 2017 – May 2021

TECHNICAL SKILLS

GenAI / Agentic AI:	LLM Agents, Multi-Agent Systems, LangGraph, LangChain, RAG, GraphRAG, Prompt Engineering, Tool / Function Calling, ReAct, Knowledge Graphs, Graph Embeddings, Vector Databases, Semantic Search
ML / NLP:	PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, Transformers, BERT, GPT, BART, spaCy, NLTK, Fine-Tuning, Model Evaluation
Data / Infra:	Python, C++ , Java, Docker, Kafka, AWS, Azure, MLflow, Databricks, OpenTelemetry, MySQL, MongoDB, Redis, Elasticsearch, Neo4J, Neptune DB, Streamlit, Plotly
Focus Areas:	Agentic AI, Generative AI, LLM Applications, MLOps, Recommendation Systems, Natural Language Processing, Time Series, Cloud Computing, Data Structures & Algorithms